



۱. $y_1 = x_1 + 2x_2$, $y_2 = x_2 - 2x_3$, $y_3 = x_3 - 2x_1 + 2x_2$ - Finite strain tensor

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$$E^* = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 4 & 0 \\ -1 & 0 & 2 \end{bmatrix}, e^* = \begin{bmatrix} 62 & 9 & 5 \\ 9 & 64 & -2 \\ 5 & -2 & 74 \end{bmatrix} \frac{1}{169}$$

Lagrangian

Eulerian

$$2. \quad \vec{a} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}$$

a) Express in Cartesian and Cylindrical

b) $v_r = 0$, $v_\theta = 2r + 4/r$ at $r=2$ → acceleration field and rate of deformation:

$$\Rightarrow \left(a_r = -4r - \frac{16}{r^3} \right)_{r=2} + a_\theta = 0, a_z = 0 \quad \text{at } r=2 \quad \Rightarrow \left(a_r = -18\hat{e}_r, a_\theta = a_z = 0 \right)$$

rate of deformation tensor: $D = 1/2 (\nabla \vec{v} + (\nabla \vec{v})^T)$

$$\left. \begin{aligned} a_r &= -18\hat{e}_r, a_\theta = a_z = 0 \\ D &= \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned} \right\}$$

$$\frac{\partial}{\partial t} y_i = \frac{f_i(R,t)}{R} x_i, \quad R^2 = x_1^2 + x_2^2 + x_3^2, \quad \text{all } i \text{ and } x \text{ values}$$



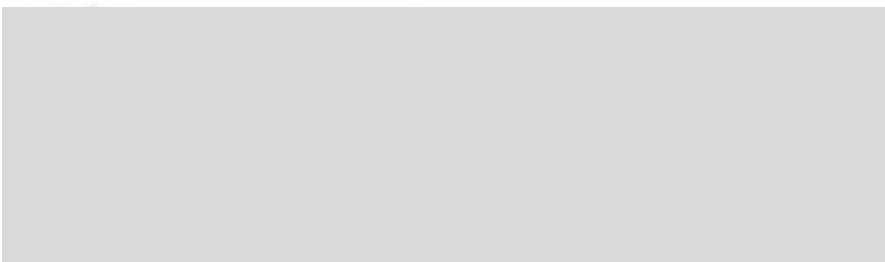
deformation

$$F = \begin{bmatrix} \lambda + \frac{1}{R} x_1^2 & \frac{\lambda_1 R}{R} x_1 x_2 & \frac{\lambda_1 R}{R} x_1 x_3 \\ \frac{\lambda_1 R}{R} x_2 x_1 & \lambda + \frac{\lambda_1 R}{R} x_2^2 & \frac{\lambda_1 R}{R} x_2 x_3 \\ \frac{\lambda_1 R}{R} x_3 x_1 & \frac{\lambda_1 R}{R} x_3 x_2 & \lambda + \frac{\lambda_1 R}{R} x_3^2 \end{bmatrix}, \quad \lambda = \frac{f(R,t)}{R}, \quad \lambda_1 R = \frac{f_R}{R} \cdot \frac{f}{R^2}$$

gradual tensor

b) find the velocity acceleration field by referential and spatial description.

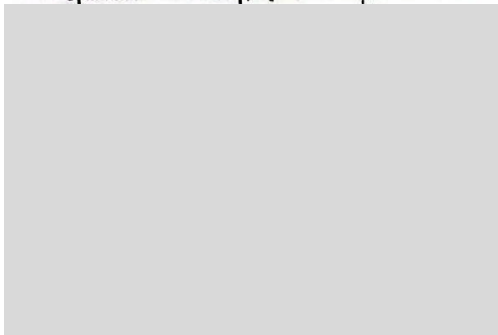
$$\mathbf{v} = \frac{\partial \mathbf{y}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{f(R,t)}{R} x_i \right)$$



$$\frac{f(R,t)}{R} x_i$$

$$\frac{f(R,t)}{R} x_i$$

Sketch description:



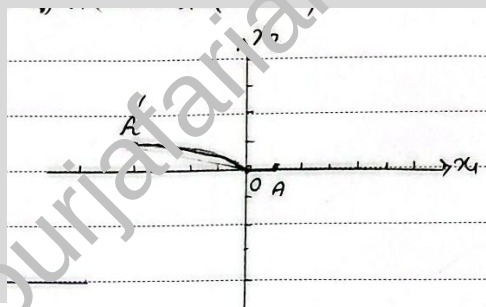
$$f(R_H) = 0 \quad r = f(R_H) \quad , \quad x_i = \frac{R}{P} y_i$$

$$v_i(y_H) = \frac{\dot{P}(R_H)}{f(R_H)} y_i \quad r = f(R_H)$$

$$a_i(y_H) = \frac{\ddot{P}(R_H)}{f(R_H)} y_i$$

$$4 \quad x_1 = x_1 - 2\beta x_2^2 t^2, \quad x_2 = x_2 - kx_3 t, \quad x_3 = x_3, \quad \beta=2, k=3$$

a) $t=0 \rightarrow O(0,0,0), A(0,1,0)$, @ $t=1$ draw OA



b) Velocity at $t=2$, @ $t=0$ at $(1,3,1)$

$$\Rightarrow (v_1, v_2, v_3) = (-144, -3, 0)$$

c) velocity of particle $(1,3,1)$ $t=2$

$$\Rightarrow (x_1, x_2, x_3) = (187, 9, 1)$$

$$\Rightarrow (v_1, v_2, v_3) = (-1296, -3, 0)$$

$$5 \quad x = X + X_1 k e_1, \quad dx^{(1)} = (ds_1 / \sqrt{2})(e_1 + e_2), \quad dx^{(2)} = (ds_2 / \sqrt{2})(-e_1 + e_2)$$

a) deformed element $dx^{(1)}, dx^{(2)}$

$$dx^{(1)} = \frac{ds_1}{\sqrt{2}} (1+k)e_1 + e_2$$

$$dx^{(2)} = \frac{ds_2}{\sqrt{2}} (-(1+k)e_1 + e_2)$$

b) stretches of elements and change angles $ds_1/dS_1, ds_2/dS_2$

$$\frac{ds_1}{dS_1} \frac{ds_2}{dS_2} = \left(\frac{1}{2} k^2 + k + 1 \right)^{1/2}$$

$$\Delta\theta = \cos^{-1} \left(\frac{-k^2 - 2k}{k^2 + 2k + 2} \right) = \pi/2$$

c) part (b) for $k=1, k=10^{-2}$

$$\frac{ds_1}{ds_1} = \frac{ds_2}{ds_2} = \sqrt{5/2} \cdot \Delta\theta = 36.87$$

$$k=1$$

$$\frac{ds_1}{ds_1} = \frac{ds_2}{ds_2} = 1.005, \Delta\theta = 0.1$$

d) compare result of (c) to that predicted by the small strain tensor E .

$$\frac{ds_1}{ds_1} = \frac{ds_2}{ds_2} = 1 + 1/2 k = 1.005 \text{ as same as c}$$

almost:

$$\Delta\theta = k \cdot 0.1 \text{ as same as c}$$

$$6 \quad x_1 = x_1 + kx_3, \quad x_2 = x_2 + kx_3, \quad x_3 = kx_3, \quad k = 10^{-5}$$

a) strain tensor components

$$E = \begin{bmatrix} 0 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 0 \end{bmatrix} \times 10^{-5}, \quad E_{13} = E_{31} = 5 \times 10^{-6}, \quad E_{22} = 10^{-5}$$

b) unit elongation of an element in direction e_1 or e_2

$$\Rightarrow \frac{\Delta L}{L} = k/2 = 5 \times 10^{-6}$$

$$7 \quad u_1 = kx_1^2, \quad u_2 = kx_2x_3, \quad u_3 = k(2x_1x_2 + x_1^2) \quad \text{max elongation, } (1, 0, 0) \text{ initial}$$

initial

$$(\delta l/L)_{\max} = \lambda_{\max}(E) = k \lambda_{\max}(A) = 2.4812 k$$

8 $v = kx_2^2 e_1$

a) rate of deformation and spin tensors.

Deformation: $D = 1/2 (\nabla v + \nabla v^T) = \begin{bmatrix} 0 & kx_2 & 0 \\ kx_2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Spin tensor: $\omega = 1/2 (\nabla v - \nabla v^T) = \begin{bmatrix} 0 & kx_2 & 0 \\ -kx_2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

b) $dx = c \, dn$, $n = (e_1 + e_2)/\sqrt{2}$, $x = 5e_1 + 3e_2$

$\frac{d(ds)}{dt} = 3k \, ds$

$$\frac{d}{dt} \mathbf{d}\mathbf{n} = d\mathbf{s}\mathbf{n}$$

a) Show $\left(\frac{D}{Dt}\right)\mathbf{n} = D\mathbf{n} + \mathbf{W}\mathbf{n} - (\mathbf{n} \cdot D\mathbf{n})\mathbf{n}$

$$\frac{D\mathbf{n}}{Dt} = D\mathbf{n} + \mathbf{W}\mathbf{n} - (\mathbf{n} \cdot D\mathbf{n})\mathbf{n}$$

b) $\mathbf{n}$ is eigenvector of D . $D\mathbf{n}/\|\mathbf{n}\| = \mathbf{W}\mathbf{n} = \mathbf{W}\mathbf{x}\mathbf{n}$

$$I_{ij} = -\epsilon_{ijk} \omega_k$$

$$\omega \times n = Wn$$

$$\frac{D}{Dt} = Wn = \omega \times n$$

$$10 \quad v = d(x_1 e_1 + x_2 e_2)$$

$$p(a) = p_0 \Rightarrow p(t) = p_0 e^{-at^2}$$

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۱) $x_1 = x_1 + 3x_2$, $x_2 = x_2$, $x_3 = x_3$

a) F.C tensors

$F = \sigma x$

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$C = F F^T$

$$\begin{pmatrix} 1 & 3 & 0 \\ 3 & 10 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

b) eigenvalues and eigenvectors of C

$\lambda_1 = \frac{1}{2}(11 + \sqrt{117})$
 $\lambda_2 = \frac{1}{2}(11 - \sqrt{117})$
 $\lambda_3 = 1$

$v_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ \frac{9 + \sqrt{117}}{6} \\ 0 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ \frac{9 - \sqrt{117}}{6} \\ 0 \end{bmatrix}$

c) U, U^{-1}

$$U = \begin{bmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} & 0 \\ \frac{3}{\sqrt{13}} & \frac{11}{\sqrt{13}} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad U^{-1} = \begin{bmatrix} \frac{11}{\sqrt{13}} & \frac{-3}{\sqrt{13}} & 0 \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$U = \frac{1}{\sqrt{13}} \begin{bmatrix} 2 & 3 & 0 \\ 3 & 11 & 0 \\ 0 & 0 & 13 \end{bmatrix}$

$U^{-1} = \frac{1}{\sqrt{13}} \begin{bmatrix} 11 & -3 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 13 \end{bmatrix}$

d) R

$$R = \begin{bmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} & 0 \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$\begin{cases} \vec{r} = (x_1 \vec{e}_1 + x_2 \vec{e}_2) / r^2 \\ r^2 = x_1^2 + x_2^2 \end{cases} \quad \begin{cases} \vec{v} = (-x_2 \vec{e}_1 + x_1 \vec{e}_2) / r^2 \\ r^2 = x_1^2 + x_2^2 \end{cases}$$

16) $V_1 = -x_2 / r^2$

$V_{1,1} = \frac{\partial}{\partial x_1} \left(\frac{-x_2}{r^2} \right) = -x_2 \left(\frac{\partial}{\partial x_1} \left(\frac{1}{r^2} \right) \right) = \frac{2x_1 x_2}{r^4}$

$V_{1,2} = \frac{\partial}{\partial x_2} \left(\frac{-x_2}{r^2} \right) = -\frac{1}{r^2} + \frac{2x_2^2}{r^4}$

۱۳. $v = \text{curl } \psi$, v is isochoric

$v = \nabla \psi$ is isochoric

$v_1 = ax_1x_3 - 2x_3$, $v_2 = -bx_2x_3$, $v_3 = 2x_1x_2$, flow is isochoric a,b relation

$$\nabla \cdot (a-b)x_3 = 0 \rightarrow a=b$$

$$a=b$$